BT4222 Interim Report

SPOTLIGHT: A Time-Aware and Diversity-Calibrated News Recommendation Framework for Homepage Slate Optimisation

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1. Machine Learning Problem

1.1 Business Objective & ML Alignment

- **Core Business Objective:** Maximise total reader engagement on premium homepage "Spotlight" slots (Top-K) for cold-start news platforms by achieving three goals
    - *Engagement Maximisation:* Identify high-performing content to maximise aggregate CTR and article reads.
    - *Dynamic Temporal Rotation:* Continuously rotate slate content to adjust for shifting reader behavior across the day, combat exposure fatigue, and replace ageing news.
    - *Freshness vs. Popularity Balance:* Surface newly published articles early based on recency signals rather than relying solely on historical cumulative clicks.

- **ML Problem Formulation:** Structured as a supervised pointwise Click-Through Rate (CTR) prediction task combined with dynamic temporal Top-K re-ranking.
    - *Probability Estimation:* Predicts P(click = 1 | u, d, t) using article metadata d (topic, freshness), user reading history u, and temporal context t (hour of day, time elapsed since publication).
    - *Dynamic Re-Ranking:* Scores and ranks candidate articles into Top-K slates at regular scheduling intervals (e.g., hourly/4-hour windows) using freshness decay functions and updated temporal/engagement features.

Business KPI	ML Evaluation Metric	Rationale
Spotlight Engagement Maximisation	*CTR@K*, *NDCG@5*, *NDCG@10*, *MRR*	• Measures whether articles that users actually click are ranked within the limited Top-K spotlight positions. • Higher values indicate that the ranking model successfully places engaging content at the top of the slate.
Temporal Rotation Efficacy	*NDCG@5*, *NDCG@10*, *MRR* on temporally held-out Day 6 data	• Evaluates whether recommendation quality is maintained across successive time windows and whether periodic

		re-ranking improves engagement relative to a static ranking baseline.
Freshness Content Promotion	*NDCG@K* and *CTR@K* on fresh articles	• Evaluates whether the model can successfully recommend newer articles, rather than only prioritising historically popular content.
Personalisation vs. Popularity Balance	*NDCG@K*, *MRR*, *AUC*	• Measures whether personalised recommendations correctly rank articles that individual users are likely to click, beyond simply recommending globally popular content.

Table 1: Business KPI to ML Metric Mapping

1.2 Model Inputs & Target Outputs

Input Sample Definition

For each impression, the model receives the user's previously clicked news history $H_u = [h_1, ..., h_P]$ and the candidate news set $C_u = [c_1, ..., c_M]$, called *impression slate*, displayed in the current impression. The model produces one click score $\hat{s}_i$ for each candidate c_i.

Model Input Feature Space

The model input feature space consists of:

- **User's history news features:** category, subcategory, title, abstract, and optional entity features of the articles in H_u.
- **Candidate news features:** category, subcategory, title, abstract, and optional entity features of each article in C_u.

The model predicts a click-score vector:

$$\hat{S}_u = [\hat{s}_1, ..., \hat{s}_M]$$

The candidate articles are ranked in descending order of their predicted click scores.

The ground-truth click labels from ***impressions*** are used as the target:

$$Y_u = [y_1, ..., y_M], y_i \in \{0, 1\}$$

where $y_i = 1$ indicates that candidate c_i was clicked, and $y_i = 0$ indicates that it was not clicked.

2. Dataset and Impression Data Understanding

2.1 Data Overview & Sampling Scheme

2.1.1 Data Source

This study uses the MIND (Microsoft News Dataset), specifically the MIND-small train and development splits released by Microsoft. MIND is a large-scale English-language news recommendation dataset containing user click logs collected from Microsoft News (MSN). The dataset consists of two linked components: a behaviours log, which records user impression and click events, and a news catalogue, which contains article metadata such as title, abstract, category, and subcategory.

2.1.2 Sampling Scheme

The official MIND-small release provides separate training and development datasets. To better align with the project's business objective of predicting future user engagement and dynamically updating homepage spotlight recommendations, the two datasets were first combined and subsequently re-partitioned using a temporal holdout strategy.

A temporal split more closely reflects a real-world deployment setting, where models are trained on historical interactions and evaluated on future user behaviour. This approach also prevents information leakage that may occur when future interaction data is used during training. The combined dataset spans seven calendar days, from 9 November 2019 to 15 November 2019, and is split chronologically into training, validation, and test partitions. Specifically, Days 1 - 4 will be for training, Day 5 for validation, and Days 6 - 7 for testing.

To verify that the temporal holdout strategy produces a representative evaluation environment, descriptive statistics are compared across the resulting training, validation, and test partitions. Specifically, impression volumes, click-through rates, unique user counts, unique article counts, and impression slate lengths are examined. These statistics are used solely to assess the representativeness of the resulting partitions, rather than to determine the split boundaries. Consistency across these measures would indicate that the model is evaluated on future interactions drawn from a similar underlying distribution while preserving chronological integrity and preventing information leakage.

Split	No. of Impressions	Percentage of Total Impressions	No. of Unique Users	No. of Unique News (Exposed)
Train	126695	55.1%	46012	16978
Validation	30270	13.2%	20179	6144
Test	73152	31.8%	50000	5369

Table 2: Descriptive Statistics Across Chronological Splits (5-1-1 Split)

Split	CTR Stability	Average Impression Slate Lengths
Train	0.041	36.474
Validation	0.038	40.384
Test	0.041	37.470

Table 3: Distributional Comparison Across Chronological Splits (5-1-1 Split)

The resulting temporal partitions exhibit broadly consistent interaction characteristics. The click-through rate (CTR) remains stable across the training (4.1%), validation (3.8%), and test (4.1%) sets, suggesting that user engagement behaviour does not undergo substantial distributional shifts throughout the seven-day period. This indicates that model performance on the validation and test sets is likely to reflect genuine predictive capability rather than changes in underlying user behaviour. Additionally, average impression slate lengths remain relatively consistent across all partitions, indicating that users are exposed to a similar number of candidate articles and that ranking complexity remains stable throughout the evaluation period.

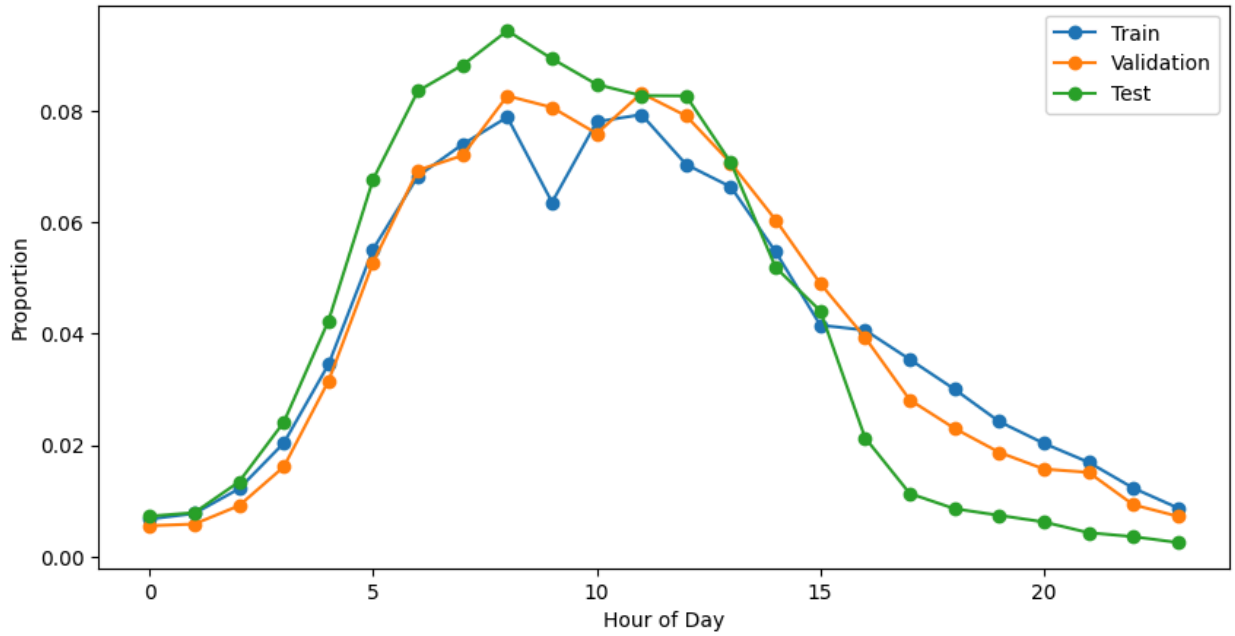


Fig.1: Hour User Activity Distribution

Hour-of-day distributions are largely consistent across the training, validation, and test sets. This suggests that the temporal split preserves diurnal user engagement patterns, and indicates that the validation and test sets capture the same broad consumption behaviours observed during training, including periods of high and low user activity throughout the day. This similarity in hourly interaction patterns is particularly important for this project, as homepage spotlight allocation and article rotation decisions are expected to vary throughout the day.

Collectively, these statistics indicate that the temporal holdout strategy produces representative training, validation, and test partitions while preserving chronological ordering.

2.1.2 Entity Counts

Total sample size and entity counts. The combined train, validation and test behaviour logs contain 230,117 user behaviour records, corresponding to 156,965 impression events generated by 94,057 unique users. These interactions involve a news catalogue containing 65,238 unique news articles.

Time span. The logged impression events span 9 November 2019 00:00:19 to 15 November 2019 23:58:03. It is worth noting that this period reflects only the impression-observation window captured by the MIND dataset. Each user's History field, which represents previously clicked news articles, may contain interactions that occurred before the observation window and does not provide corresponding timestamps. Consequently, this study distinguishes between the impression-observation window used for temporal modelling and evaluation, and the historical reading profile used as a proxy for long-term user interests.

2.2 Impression, Exposure & Interaction Dynamics

The impressions dataset was derived by merging the behaviors dataset for all 7 days and expanding each impression slate into one row per exposed article. This resulted in 8,584,442 exposures recorded.

2.2.1 Impression Slate & Position Distribution

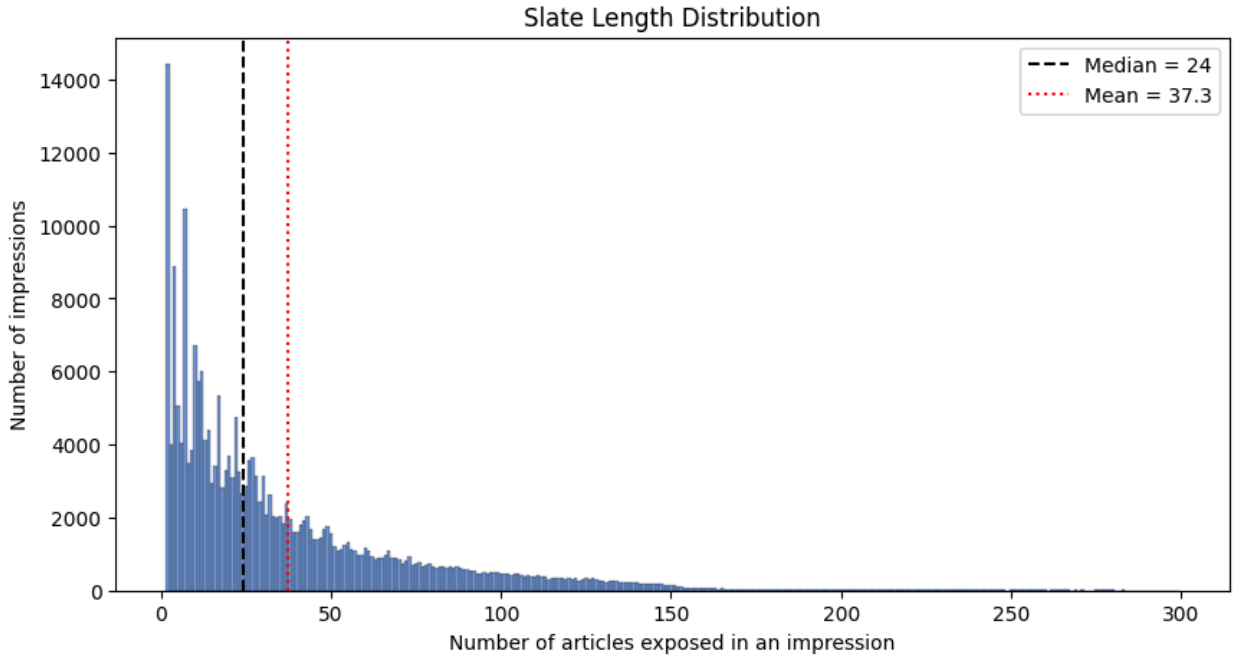


Fig.2: Slate Length Distribution

Slate Length. Slate length is right-skewed (Fig.2), with a mean of 37.3 articles, a median of 24, an interquartile range of 10–51, and a range of 2–299 articles. As the mean is higher than the median, it indicates that a small number of impressions have an unusually high slate length. At least 25% of impressions contain 10 or fewer exposed articles, whereas half of the impressions contain between 10 and 51 articles.

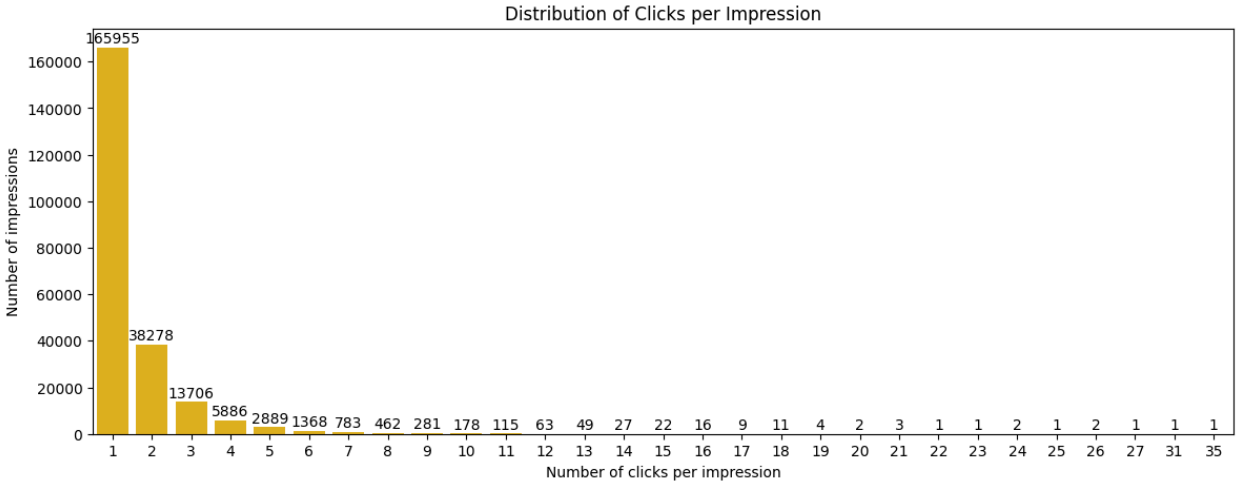


Fig.3: Click Distribution per Impression

Impression-level clicks. Most impressions (72.1%) receive exactly one click, while the remaining 27.9% receive multiple clicks (Fig.3). No impressions contain zero clicks in this MIND-small dataset, and the maximum number of clicks per impression is 35. This indicates that user interactions are concentrated around a single clicked article for most impressions, with a smaller proportion of impressions generating multiple clicks. Since the dataset contains no zero-click impressions, impression-level click-through analysis is limited. Therefore, subsequent CTR analysis focuses on clicks at the article exposure level.

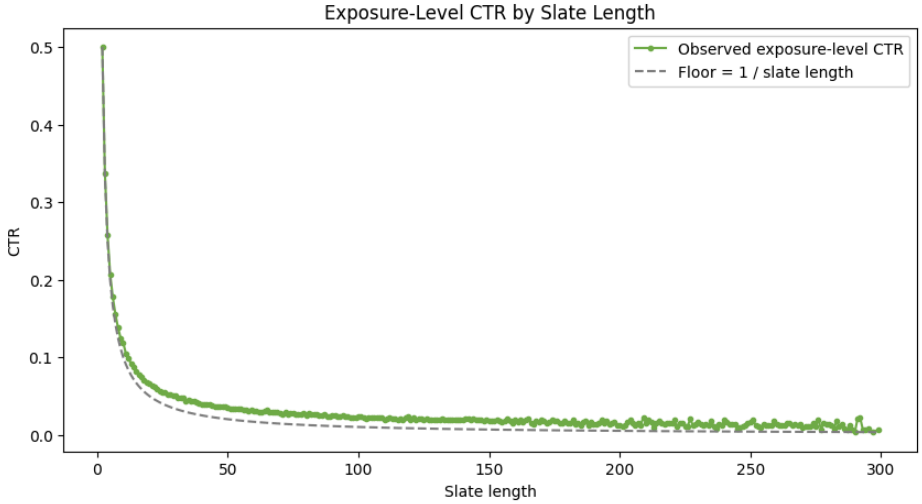


Fig.4: Exposure-level CTR by Slate Length

Exposure-level CTR across slate length. Since each impression contains at least one click, the exposure-level CTR has a lower bound of 1/L, where L is the slate length. The CTR decreases from 50.0% for slates with 2 articles to below 1% for slates approaching 300 articles (Fig.4). This pattern is expected because clicks are distributed across a larger number of exposures when slate length increases. Most observed CTR are close to the lower bound of 1/L, indicating that many impressions receive only one click. Therefore, the decreasing CTR should not be interpreted as evidence that users are less likely to click when shown longer slates.

Exposure-level CTR across positions/display ranks. Based on the MIND documentation, the news articles within each impression were shuffled. Hence, the article order in the dataset does not represent the actual display ranking. Therefore, CTR by display rank cannot be meaningfully analysed as the recorded positions are arbitrary.

2.2.2 Interaction Hierarchy & Funnel

The observable interaction hierarchy in MIND-small is limited to article exposure → click, as the dataset does not record downstream behaviours such as reading time, article completion etc.

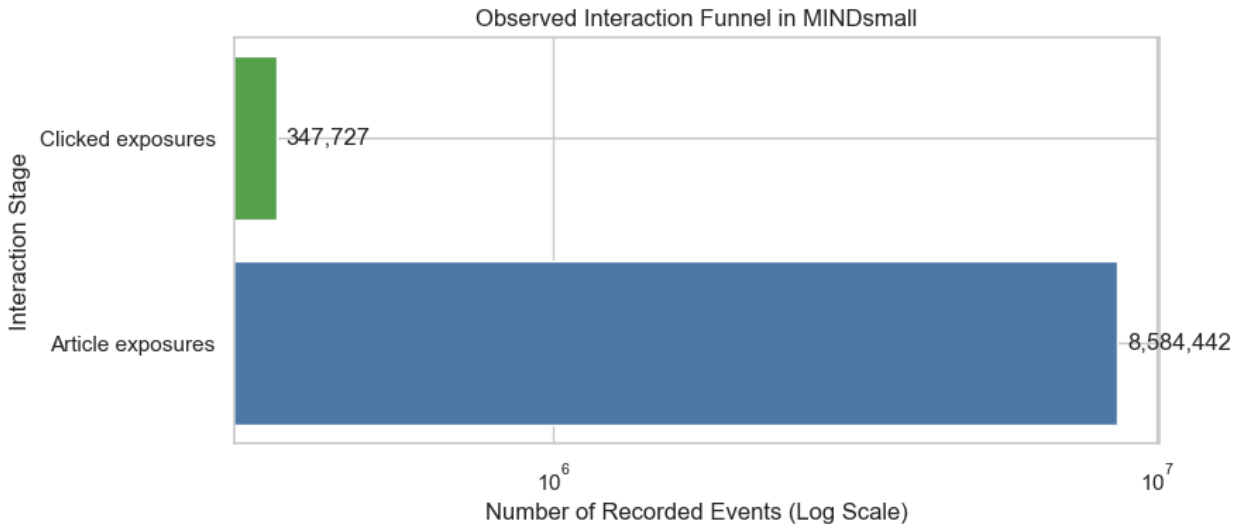


Fig.5: Observed Exposure-to-click Interaction Funnel

Across 230,117 impressions, a total of 8,584,442 article exposures were observed, of which 347,727 resulted in clicks (Fig.5). This corresponds to an exposure-level CTR of 4.05%, while the remaining 8,236,715 exposures (95.95%) were exposed but not clicked.

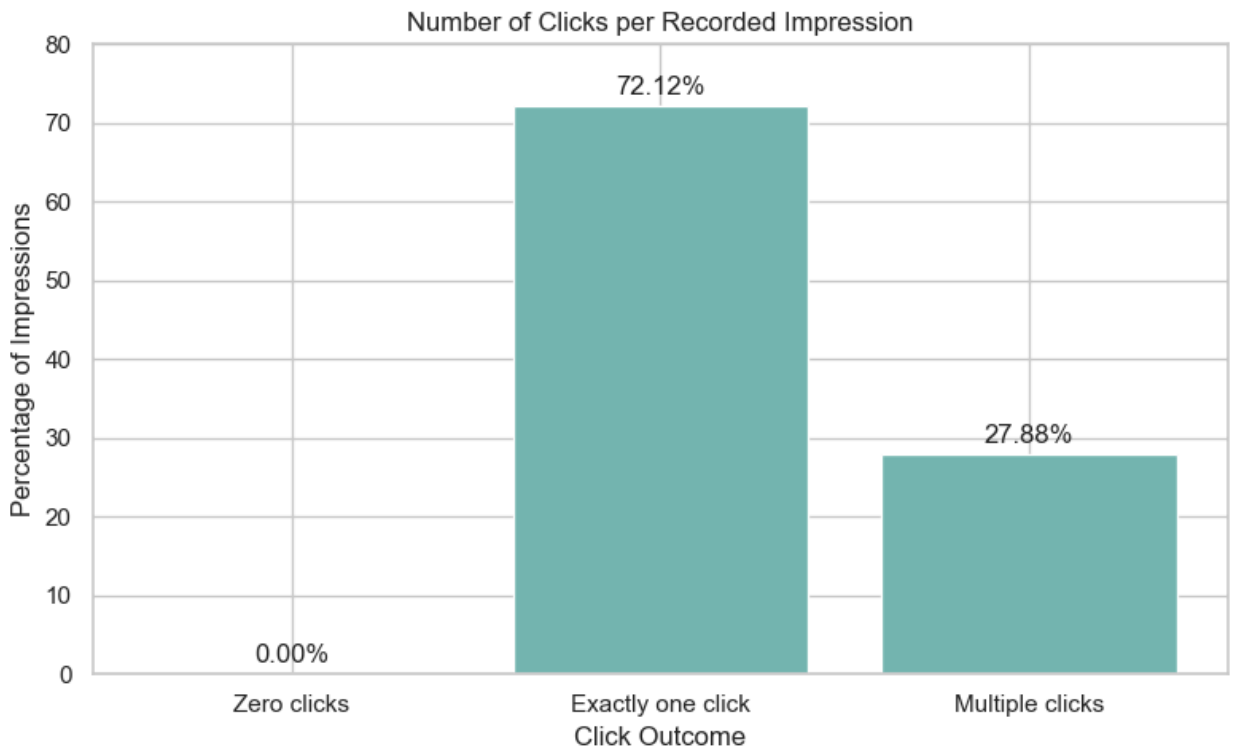


Fig.6: Observed Absence of Zero Clicks

At the impression level, all 230,117 impressions records contain at least one click. However, this should not be interpreted as a real-world conversion rate of 100%, as zero-clicked impressions are absent from the training subset (Fig.6). The observed funnel therefore supports analysis of which exposed articles were selected within a recorded impression, but cannot estimate whether an arbitrary homepage slate will generate any engagement. Similarly, the lack of post-click signals prevents analysis of whether a click resulted in sustained reading or other higher-value engagement.

2.2.3 Feedback Semantics

In MIND, feedback is recorded at the article-exposure level. An exposed article is assigned a label of 1 if it was clicked and 0 if it was displayed within the same impression but not clicked. Articles absent from a user's recorded impressions are unobserved rather than negative examples because there is no evidence that the user saw them.

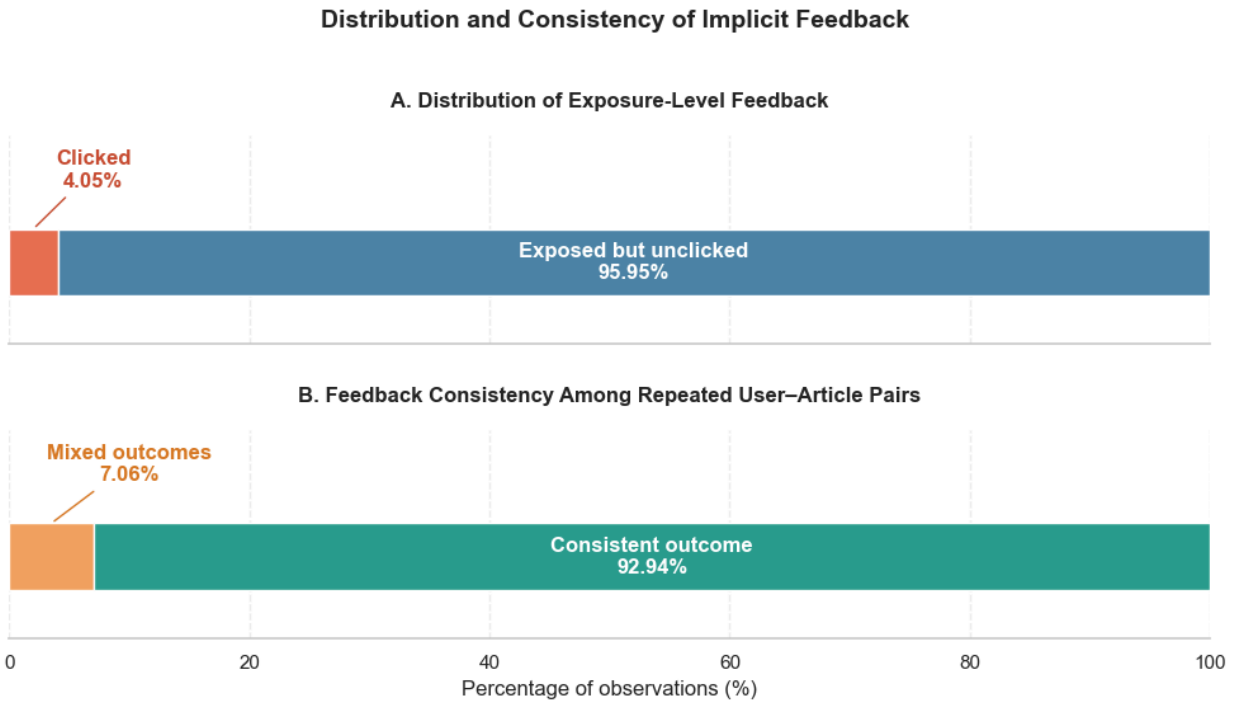


Fig.7: Distribution and Consistency of Implicit Feedback

Distribution of exposure-level feedback. Of the 8,584,442 article exposures, 347,727 (4.05%) were clicked, while 8,236,715 (95.95%) were exposed but unclicked (Fig.7, Panal A). This reveals substantial label imbalance, as unclicked exposures outnumbered clicked exposures by approximately 24 to 1. Consequently, evaluation using accuracy alone may be misleading because a model that predominantly predicts non-clicks could achieve high accuracy without identifying articles that users are likely to click.

However, an exposed-but-unclicked article should be interpreted as an implicit and relatively weak negative signal rather than a definitive evidence of dislike. A user may not click an article because of reasons such as attention span, or not having enough time to read through another exposed article etc.

Consistency of repeated user-article feedback. The merged dataset contains 7,437,956 unique exposed user–article pairs, of which 890,387 were exposed more than once. Among these repeatedly exposed pairs, 62,828 (7.06%) of them received both clicked and unclicked labels across different impressions, while the remaining 92.94% had consistent recorded outcomes (Fig.7, Panel B).

The mixed-label cases demonstrated that the same user may ignore an article during one impression but click it on another occasion. Therefore, a label of 0 does not necessarily represent a stable negative preference. Click outcomes may depend on contextual factors such as exposure timing, competing articles and changes in the user's immediate interests.

Overall, MIND distinguishes 3 feedback states:

1. Clicked exposures: Articles appear in the impression with a label 1.
2. Exposed but unclicked: Articles appear in the impression with a label 0.

3. Unobserved articles: Articles that do not appear in the user's recorded impression.

Clicked articles provide relatively strong positive signals. Exposed but unclicked articles provide weaker and potentially noisy negative signals. Unobserved articles should not be automatically assigned negative labels because the user may never have an opportunity to view them.

2.2.4 Sparsity and Temporal Patterns

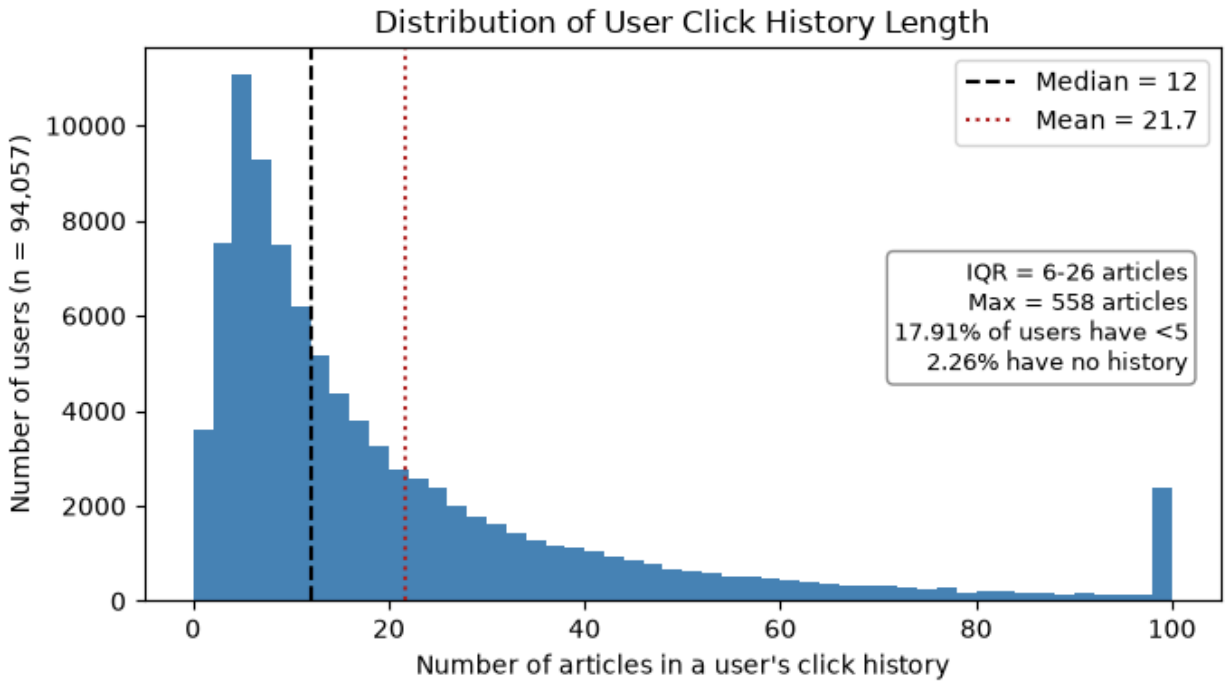


Fig.8: Histogram of User's Click History Length

User Sparsity – User's Click History. The number of articles in a user's click history is heavily right-skewed as seen in Fig.8. The mean is 21.7 articles, the median is 12 articles and the maximum history being 558 articles. 17.91%, which is nearly one-fifth of the users have fewer than 5 articles in their history, with 2.26% of users having no history at all. This substantial proportion of users provides us with limited information to personalise on.

Given that a huge bulk of the user history distribution sits in a narrow range close to 0, only a thin tail of users have enough history to describe the interests in detail. This matters because a user or article ID embedding can only be learned from that user's or article's own clicks. Thus, for most users, these embeddings would be trained on a small handful of examples and will therefore carry little reliable signal. Hence, we should explore features that do not depend on users having a long history, such as article content or time of day, since these features are shared across all users and articles.

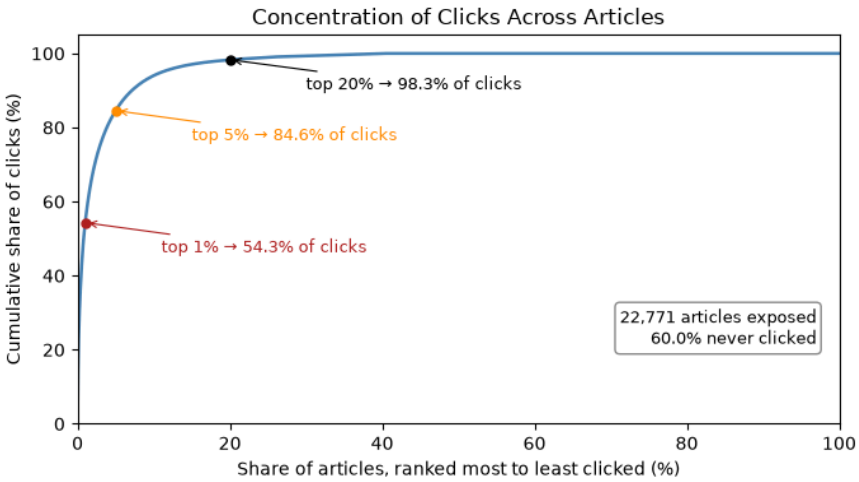


Fig.9: Cumulative Curve of Article Popularity

Article Popularity. From Fig.9, it shows that clicks are concentrated on a very small number of articles. Out of the 22,771 articles exposed minimally once, 60% were never clicked, thus leaving only around 9100 articles with positive feedback. The cumulative curve in Fig.9 rises almost vertically at the start before flattening out.

Just a mere 1% of articles account for 54.3% of all clicks, 5% of articles accounting for 84.6% of clicks and 20% of articles accounting for 98.3% of clicks. This is an extreme example of pareto's rule, where a small number of articles are responsible for a disproportionate amount of clicks.

Thus, with 80% of the articles accounting for less than 2% of the total clicks, the click signal we can learn from is limited to a small set of articles. The remaining articles were mostly never clicked at all, so the data cannot tell us whether users actually disliked the articles or these articles were simply not shown enough to gather clicks. Ultimately, article popularity not only reflects user preferences but also reflects how the platform exposed the article.

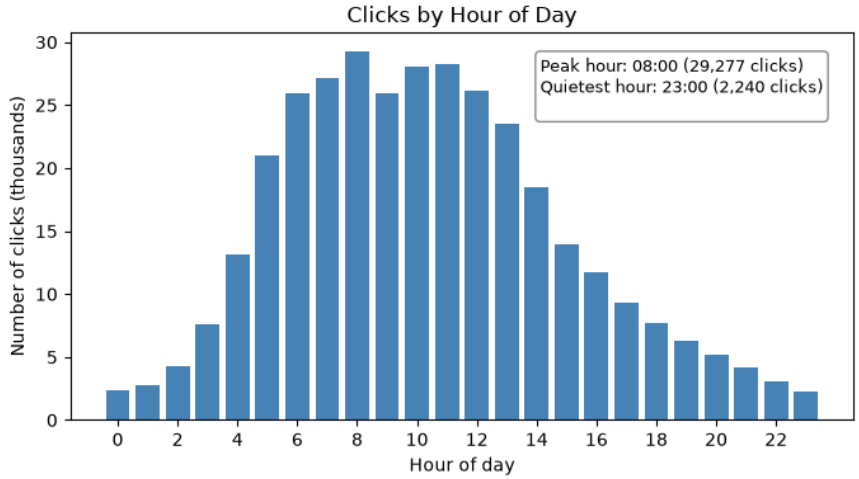


Fig.10: Clicks by Hour of Day

Click Count by Time of Day. From Fig.10, it shows that click volume follows a clear daily cycle. Activity is lowest overnight, with the quietest hours being between 11pm to 1am the next day. From 4am onwards, it rises sharply to a peak of 29,277 clicks between 8 to 9 am. This tells us that activity is heavily concentrated in the morning and early afternoon. Thus, this means that the value of a spotlight slot depends heavily on when it is filled, since the same article shown overnight would reach only a small fraction of the audience it would reach during the peak morning hours.

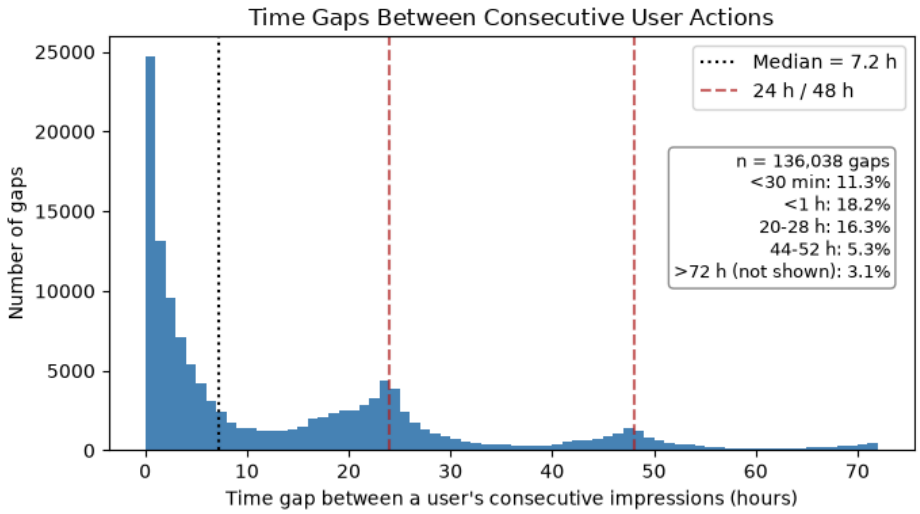


Fig.11: Temporal Gap Between User's Consecutive Sessions

Temporal Gap between User Impressions. The time between each user's consecutive impressions gives 136,038 gaps and the distribution has two clear parts.

The first is a large spike in the opening few hours, which comes from users who return within the same session or later in the day. The second is a bump at around 24 hours, with a subsequent smaller repeat at around 48 hours, showing that many users come back at roughly the same time on the next day or the day after.

Since the distribution has multiple peaks, the median gap of 7.2 hours actually falls in the quiet stretch between the peaks and does not describe a typical user well. Thus, a single average time gap is not a useful summary. The more meaningful pattern is the daily rhythm of when users return.

2.3 Feature Dictionary & Descriptive Statistics

Following our empirical findings on slate distributions, extreme popularity skew, feedback sparsity, and diurnal cycles (Section 2.2), this section establishes the comprehensive feature space engineered to power our recommendation pipeline. Rather than accumulating an unmanageable quantity of marginal features, we deliberately engineered a streamlined set of core features organized into four business-aligned domains:

1. Content Semantics: Dense text representations, knowledge graph grounding, and categorical taxonomy to solve article cold-start.
2. Freshness & Popularity Dynamics: Continuous recency decay and historical engagement conversion to balance viral content against breaking news.
3. User Reading Profile & State: Sequential reading embeddings, topical diversity entropy, category calibration, and cold-start fallback gates.
4. Temporal & Circadian Dynamics: Hour-of-day clock cycles, day-of-week context, and inactivity gaps governing dynamic spotlight slate rotation.

All descriptive statistics below are computed across the complete merged dataset combining train, validation and test splits of MIND-small (230,117 impression sessions, 94,057 unique users, 65,238 unique articles, and 8,584,442 total exposures). Crucially, in strict adherence to data hygiene protocols, all features are computed using only interactions preceding session timestamp t, preventing future data leakage into validation and test evaluation.

2.3.1 Feature Metadata Dictionary

Table 4 details the metadata for all engineered features, including their description, data type, representation dimension/unit, and source table.

Feature Name	**Feature Type**	**Domain**	**Data Type / Unit**	**Source Table**
article_content_emb	Text / Corpus & Embedding	Content Semantics	Dense Vector (384)	news.tsv
Description	An article embedding vector representing the semantic meaning of the article.			
Justification	Article content embeddings capture the semantic content of news articles and enable the model to identify topical relevance between readers and candidate stories. This supports the business objective of maximising spotlight engagement by promoting articles that align with audience interests, even when limited historical interaction data exists.			
kg_entity_emb	Knowledge Graph & Embedding	Content Semantics	Dense Vector (100)	news.tsv & entity_embedding.vec
Description	Average 100-dimensional TransE entity embeddings of linked Wikidata knowledge graph entities identified in the title and abstract.			
Justification	Knowledge graph entity embeddings encode relationships among people,			

	organisations, locations, and events mentioned within articles. These representations help identify relevant breaking stories and support the timely promotion of fresh content, reducing reliance on historical popularity signals alone.			
category_id	Categorical	Content Semantics	18 categories / 16-dim embedding	news.tsv
Description	Top-level editorial category of an article.			
Justification	Article category information enables the model to capture category-specific engagement patterns and diurnal consumption trends. This is particularly important for dynamic spotlight rotation, as user preferences may shift across time periods, such as increased consumption of business news during daytime hours and sports or entertainment content during evening periods.			
subcategory_id	Categorical	Content Semantics	270 categories / 32-dim embedding	news.tsv
Description	Fine-grained editorial subcategory of an article.			
Justification	Captures more specific reader interests within broad editorial categories.			
article_recency	Numeric	Freshness & Popularity	Continuous (hours)	behaviors.tsv
Description	Elapsed time (in hours) between an article's initial publication/appearance in logs and the current impression session timestamp t.			
Justification	*article_recency* supports freshness-popularity balance and dynamic temporal rotation as it provides an anti-stagnation mechanism that allows fresher news to overcome the initial click deficit against the older and established articles, ensuring that the homepage recommendations remain dynamic and fresh. The time that a news article is recommended on the spotlight slot is critical to its engagement as user engagement is heavily concentrated during morning surges between 8–9am (Fig.10). *article_recency* ensures that high-traffic morning windows are populated with newly published morning news rather			

	than decaying news from the previous day.			
historical_ctr	Numeric	Freshness & Popularity	Continuous [0, 1]	behaviors.tsv
Description	Empirical click-through rate at the article exposure level, calculated strictly from historical exposures prior to session timestamp t.			
Justification	Article clicks follow an extreme Pareto law (1% of articles drive 54.3% of clicks). Historical CTR provides a strong engagement baseline. However, as 60% of articles have zero clicks, it must be paired with recency to avoid permanently penalizing newly published articles.			
user_history_emb	User Profile & Embedding	User Profile	Dense Vector (384)	behaviors.tsv & news.tsv
Description	Multi-head attention-weighted average of content embeddings from the user's previously clicked articles prior to session timestamp t.			
Justification	*user_history_emb* allows recommendations to move beyond generic global popularity by tailoring spotlight candidates to an individual reader's specific preferences. *user_history_emb* represents the user reading history *u* in dense vector space and is used to compute similarity against candidate article content embedding (article_content_emb) to better model *P(click = 1 | u, d, t)* for recommendations. *user_history_emb* can help to address sparsity issues. As distribution of user click history length is heavily right-skewed (Fig.8) with 17.91% of users having fewer than 5 clicked articles and 2.26% having zero history, user_history_emb helps to map sparse user's few clicked articles into continuous dense vectors to generalize across shared topic spaces even with minimal interaction history. *user_history_emb* also helps to surface the single most engaging article to the top of an impression slate. As impression slates are large (mean of 37.3 articles per slate, max 299), a single impression generally contains multiple globally popular news articles. However, as 72.1% of impressions yield only a single click, users are highly selective and will ignore the remaining 36+ recommendations. Hence, selecting the single top recommendation using global popularity alone is insufficient as it assigns uniformly high scores to generic viral articles without any individual signal to break ties. Matching candidate			

	article content embeddings (*article_content_emb*) against *user_history_emb* provides the granular and individual signal required to surface the single most engaging article to the top of a 37-article slate.			
category_affinity	Continuous Vector	User Profile & Interaction	Continuous Vector (18)	behaviors.tsv & news.tsv
Description	Normalized 18-dim frequency vector of historical category reads (category_affinity)			
Justification	*category_affinity* captures how strongly each user prefers out of the different topics, so Spotlight slots can be personalized instead of showing the same popular articles to everyone. An article's category is known at publish time, so new articles with no click history can still be scored for relevance.			
category_match	Binary categorical	User Reading Profile	Binary {0, 1}	behaviors.tsv, news.tsv
Description	Binary flag indicating match with user's top category. Indicates whether a candidate belongs to the user's historically most-read category.			
Justification	*category_match* flags whether a candidate article is the user's most read category, which is a strong predictor of a click. As it only uses article category, it works for newly published articles with no click history which supports cold-start ranking. However, when used alone, it risks a filter bubble so it should be paired with *reading_entropy* to control how strongly it is applied per user.			
reading_entropy	Numeric	User Profile	Continuous [0, ln(18) ~ 2.89]	behaviors.tsv & news.tsv
Description	Shannon entropy of user's historical category distribution. It measures topical breadth vs. specialization.			
Justification	Measures how broadly a user reads, which tells the model how much to trust category signals. Users with lower entropy should be recommended articles that are aligned with their preferred topics, while users with higher entropy respond better to diverse articles with more fresh content.			

history_length	Numeric	User Reading Profile	Integer count	behaviors.tsv
Description	Number of historical clicked articles available for the user prior to the current impression.			
Justification	*history_length* indicates how reliable the user's profile is. Features such as *category_affinity* and *reading_entropy* built from a small number of clicks are noisy, as compared to features being built from many clicks. This allows the model to weigh personalisation strongly for users with richer history and lean on general article popularity for lighter users. Conversely, this feature is also able to navigate the model when to handle new cold-start users.			
has_history	Binary categorical	User Reading Profile	Binary {0, 1}	behaviors.tsv
Description	Indicates whether the user has at least one historical clicked article.			
Justification	For new users, the category features are all zero. By clearly indicating whether the user has history, it prevents the model from misreading it as a real preference signal. *has_history* tells the model to ignore personalisation signals for such users with no history and fall back on general article popularity and freshness.			
session_hour	Temporal / Categorical	Temporal Dynamics	Integer / Cyclic [0, 23] (16-emb)	behaviors.tsv
Description	Hour of the day (0 ... 23) in local time when the impression occurred.			
Justification	Captures within-day variation in reader activity to support dynamic temporal rotation of spotlight content. EDA shows a clear diurnal activity cycle, with peaks at around 8-9am. Encoding session hour therefore provide the model with temporal context *t* when estimating *P(click = 1 | u, d, t)*. This would allow the article rankings to adapt across different periods of the day rather than remaining static.			

session_dow	Categorical	Temporal Dynamics	Categorical [0, 6] (8-emb)	behaviors.tsv
Description	Day of the week (*0 = Monday ... 6 = Sunday*).			
Justification	Provides a broader temporal context beyond hour of day, allowing the model to account for potential reader activity and content demands across days. This feature supports the project's dynamic temporal rotation objective by allowing rank decisions to condition on both the current day and current hour. However, as MIND-small covers only 7 days, this feature is treated as contextual information rather than evidence of persistent weekly behaviour trends.			
inactivity_gap	Numeric & Categorical	Temporal Dynamics	Continuous (hours) & 4-Class Bins	behaviors.tsv
Description	Continuous hours elapsed since previous recorded impression (time_since_last_click), binned into <1h, 1h–6h, 6h–24h, >=24h.			
Justification	Captures the user's current temporal state rather than absolute clock time. EDA shows substantial variation in return intervals, with a median gap of 7.2 hours. Concentrations of returning activity within short intervals and around 24-hour intervals were also shown in EDA. This helps distinguish users continuing their recent activity from those returning after longer inactivity, allowing our spotlight rankings to adapt to changes in the recency of observed user behaviour.			
slate_length	Numeric	Session Context	Integer [2, 299]	behaviors.tsv
Description	Total number of candidate news articles exposed in the current impression slate.			
Justification	Captures the competitive context of each impression, as an article competes with a varying number of alternatives for user attention. EDA shows substantial variation in slate size, with a median of 24, mean of 37.3 and a large range of 2-299 articles. Since each recorded impression contains at least 1 click, exposure-level CTR decreases as slate length increases. Thus, including slate_length provides context on the number of			

	competing articles when estimating click probability.

Table 4: Feature Metadata Dictionary

2.3.2 Numeric Features & Descriptive Statistics

Table 5 reports descriptive statistics for the ***scalar numeric features*** defined in Table 4. In accordance with the interim report requirements, *minimum, maximum, mean, median, standard deviation, and missing-value counts* are reported.

Feature Name	Count (N)	Mean ± Std Dev	Median [Q1, Q3]	Min	Max	Missing (%)
slate_length (sessions)	230,117	37.30 ± 38.90	24.00 [10.00, 51.00]	2.00	299.00	0 (0.00%)
history_length (user-level)	94,057	21.66 ± 27.78	12.00 [6.00, 26.00]	0.00	558.00	2,122 (2.26%)*
history_length (session-level)	230,117	32.46 ± 39.91	19.00 [8.00, 42.00]	0.00	558.00	5,452 (2.37%)*
historical_ctr (articles)	22,771	0.0432 ± 0.1165	0.00 [0.0000, 0.0356]	0.00	1.00	0 (0.00%)
article_recency (hours)	200,000**	23.42 ± 23.14	16.34 [10.00, 26.08]	0.00	143.48	0 (0.00%)
inactivity_gap (hours)	136,060	16.70 ± 21.04	7.24 [1.67, 23.86]	0.00	165.28	94,057 (40.87%)***
reading_entropy (user)	91,935	1.43 ± 0.51	1.50 [1.10, 1.82]	0.00	2.54	2,122 (2.26%)****

Table 5: Descriptive Statistics for Numeric Features

Note:
* A zero *history_length* count represents a valid cold-start state rather than an unavailable observation.
** *article_recency* statistics were estimated from a sample of 200,000 logged exposure events.
*** *inactivity_gap* is undefined for the first recorded impression of each user because no previous session exists.
**** *reading_entropy* cannot be calculated for users without historical interactions.

Moreover, vector representation of *category_affinity, article_content_emb, kg_entity_emb, and user_history_emb* are not summarized using scalar descriptive statistics because individual vector dimensions do not have directly interpretable units.

2.3.3 Categorical Features & Descriptive Statistics

Table 6 summarises the categorical features defined in Table 4. For each feature, the number of distinct categories, representative values and their frequencies, and missing-value counts are reported.

Feature Name	Cardinality	Missing (%)	Planned Representation
category_id	18	0 (0.00%)	Learned 16-dim embedding
subcategory_id	270	0 (0.00%)	Learned 32-dim embedding
session_hour	24	0 (0.00%)	Cyclic encoding / learned embedding
session_dow	7	0 (0.00%)	Learned embedding / one-hot
inactivity_bin	4 + new-user state	94,057 (40.87%)*	One-hot / learned embedding with [NEW_USER] state
category_match	2	0 (0.00%)	Binary
has_history	2	0 (0.00%)	Binary

Table 6: Descriptive Statistics for Categorical Features

Note:

* The missing *inactivity_bin* observations correspond to each user's first recorded impression and are assigned a dedicated [NEW_USER] state during modelling.

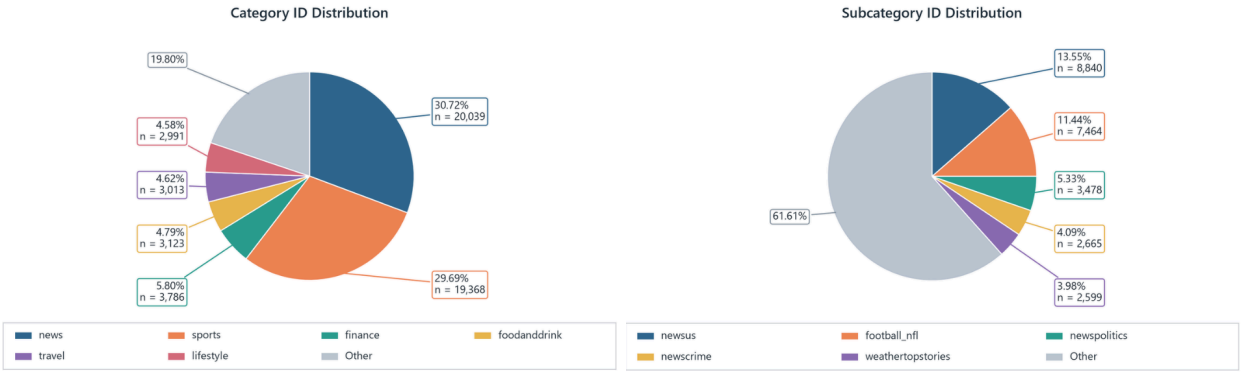


Fig.12: Distributions for category_id and subcategory_id

The *category_id* distribution is strongly concentrated: news and sports together account for 60.41% of catalogue articles. The much larger number of subcategories (270) provides finer topical distinctions that can support more specific personalization.

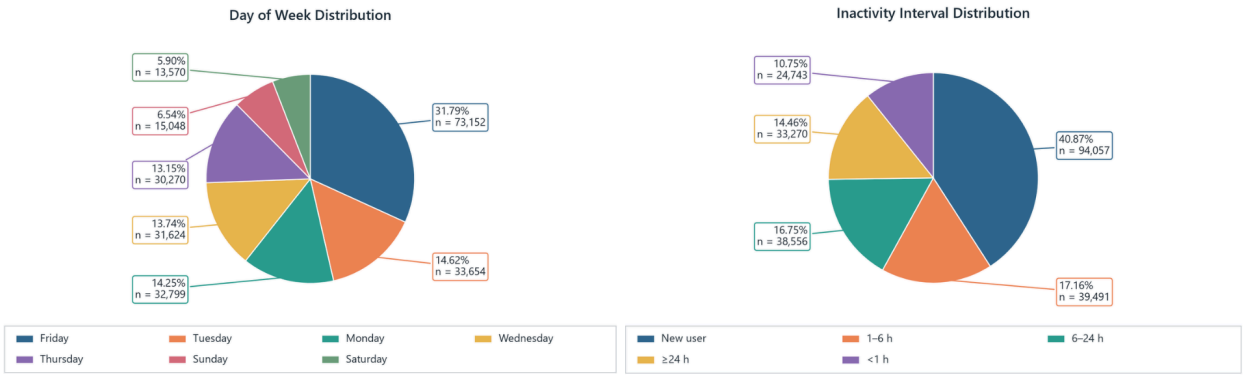


Fig.13: Distributions for session_dow and inactivity_bin

Temporal categorical variables further reveal substantial variation across the observation window. *session_hour* captures within-day behavior, while *session_dow* captures differences across the seven-day cycle. *inactivity_bin* complements these absolute temporal features by indicating whether a user is continuing a recent browsing episode or returning after a longer absence.

———————

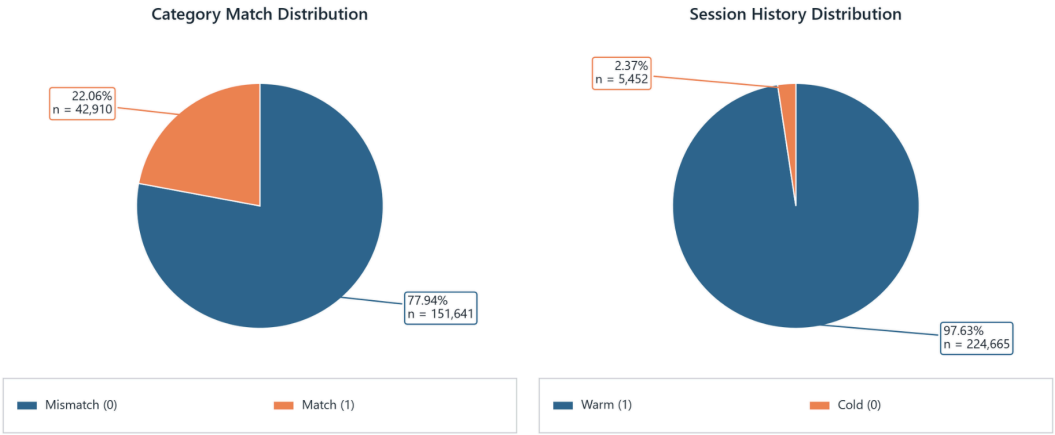


Fig.14: Distributions for category_match and has_history

The *category_match* distribution also shows that only 22.06% of evaluated candidate exposures belong to a user's historically dominant category. Together with the observed variation in *reading_entropy*, this provides useful information for calibrating recommendation diversity rather than restricting recommendations to a single dominant interest.

2.3.4 Text / Corpus & Knowledge Graph Statistics

The principal non-tabular content features in Table 4 are *article_content_emb* and *kg_entity_emb*. Before these representations are generated, their underlying raw text and entity inputs are characterized in Table 7. The project guidelines specifically recommend reporting text-length distributions and vocabulary sizes for text/corpus data.

Text / Entity Attribute	Related Feature	Median [Q1; Q3]	Mean	Range [Min; Max]	Std. Dev	Corpus / Entity Coverage	Missing (%)
Title Word Count	*article_content_emb*	10 [9; 13]	10.74	[1; 57]	3.25	728,856 tokens / 34,299 unique words	0 (0.00%)
Title Character Length	*article_content_emb*	64 [54; 78]	66.28	[11; 333]	19.13	-	0 (0.00%)
Abstract Word Count	*article_content_emb*	26 [17; 65]	36.79	[1; 474]	26.03	2,352,371 tokens / 56,577 unique words	3,415 (5.23%)
Abstract Character Length	*article_content_emb*	150 [101; 395]	219.69	[1; 2,603]	154.74	-	3,415 (5.23%)
Combined Text Vocabulary	*article_content_emb*	-	-	-	-	3,081,227 tokens / 61,000 unique words	-
Title KG Entities	*kg_entity_emb*	2 [1; 3]	2.14	[0; 18]	1.76	26,904 precomputed TransE vectors	17,543 (26.89%)
Abstract KG Entities	*kg_entity_emb*	1 [0; 3]	1.89	[0; 24]	1.95	100-dimensional TransE representations	17,522 (26.86%)
Total KG Entities / Article	*kg_entity_emb*	3 [1; 4]	3.08	[0; 32]	2.35	57,051 articles (87.45%) contain ≥1 linked entity	8,187 (12.55%)

Table 7: Text, Corpus & Knowledge Graph Descriptive Statistics

Data Imputation & Processing Conventions:

1. The text corpus consists of relatively short news content. Titles contain a median of 10 words, while abstracts contain a median of 26 words. This motivates using pretrained semantic

representations rather than relying solely on sparse lexical overlap between user histories and candidate articles.

2. For the 3,415 articles (5.23%) with missing abstracts, the title is duplicated into the abstract field prior to text encoding. This provides a non-empty textual representation while retaining the available headline semantics.
3. Knowledge graph coverage is also substantial: 57,051 articles (87.45%) contain at least one linked entity, with a median of three entities per article. For the 8,187 articles (12.55%) without linked entities, *kg_entity_emb* is initialized as a zero vector $0 \in \mathbb{R}100$, allowing the recommendation model to rely on the article's textual representation when knowledge graph information is unavailable.